MILITARY STANDARD

MIL-STD-1320-232B
(NAVY)
27 OCTOBER 1992
SUPERSEDING
MIL-STD-1320-232A (NAVY)
2 OCTOBER 1989

TRUCKLOADING

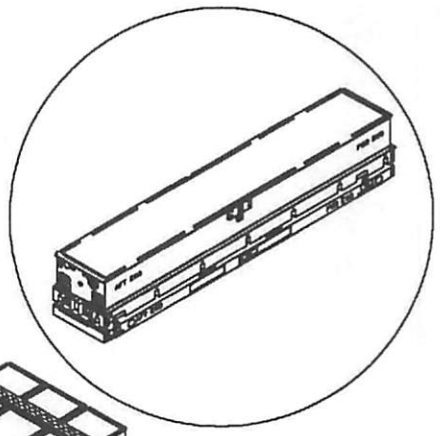
GUIDED MISSILE, AGM-119 B (PENGUIN)
IN CONTAINER, SHIPPING AND STORAGE, CNU-443A/E
WITH WINGS AND CANARDS
IN CONTAINER, SHIPPING AND STORAGE, CNU-450/E

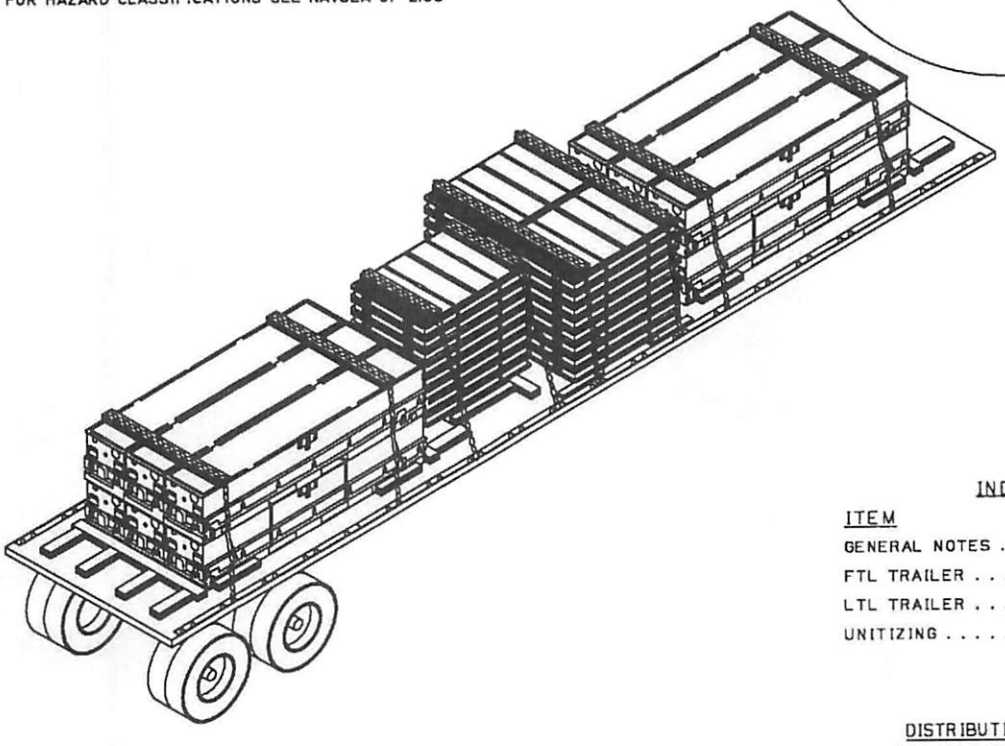
CONTAINER DATA

	CNU-443A/E	CNU-450/E
CONTAINER DRAWING	30003-3294AS100	30003-2006AS100
DIMENSIONS	134 3/4 L X 22 7/8 W X 24 3/8 H	62 1/8 L X 43 7/8 W X 7 1/8 H
GROSS WEIGHT	1231 LBS Δ	188 LBS Δ
TARE WEIGHT	468 LBS Δ	139 LBS Δ
CUBE	44.4 CU FT	11.2 CU FT

Δ DO NOT USE FOR SHIPPING WEIGHT

FOR HAZARD CLASSIFICATIONS SEE NAVSEA OP 2165





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AMSC NO. - N/A

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REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
B	REVISED EXTERNAL FEATURES OF CNU-443A/E & CHANGED CLEAT PATTERN	10/4/89	CR	Lee
A	REVISED MIXED LOADING	10-2-89	S/ER	S/KHZ
			APPROVAL	

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

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S/E.RINALDI 11-15-88
SIGNATURE TDA, WPNSTA EARLE DATE
S/L.R.CARINGI 11-18-88
SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR S/G.BENDER 10-31-88
DATE

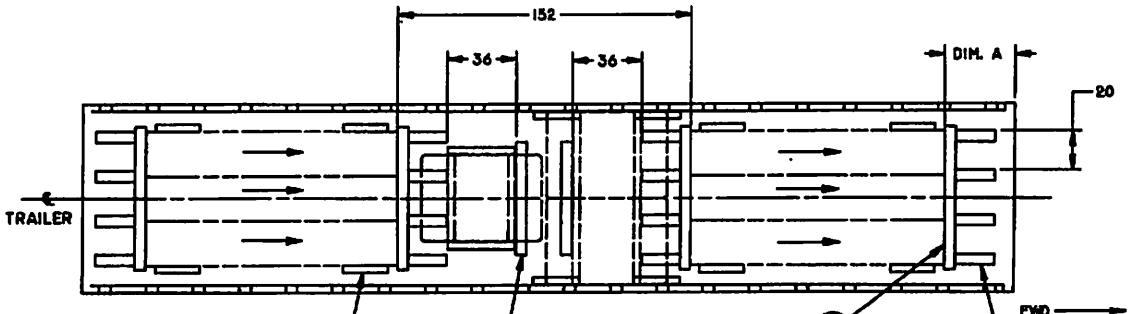
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING PENGUIN MISSILES IN SHIPPING AND STORAGE CONTAINER CNU-443A/E AND WINGS AND CANARDS IN SHIPPING AND STORAGE CONTAINER CNU-450/E.
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 40 FT THROUGH 48 FT FLATBED TRAILERS. DO NOT USE TRAILERS WITH METAL FLOORS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE TO ONE BASIS FOR TIE DOWNS.
4. STEEL STRAPPING USED AS TIE DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIE DOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAINS, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIE DOWN.
7. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION, REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
8. UNLESS OTHERWISE SPECIFIED DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER THE BLOCKING, BRACING AND TIE DOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIE DOWNS.
11. AFTER THE TARPAULINS ARE IN PLACE, ATTACH THE PROPER EXPLOSIVES PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.
12. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40 FT THROUGH 48 FT TRAILER (FLATBED)



DETAIL A

TRAILER LENGTH	DIMENSION A
40 FT	36
42 FT	48
44 FT	60
45 FT	66
48 FT	84

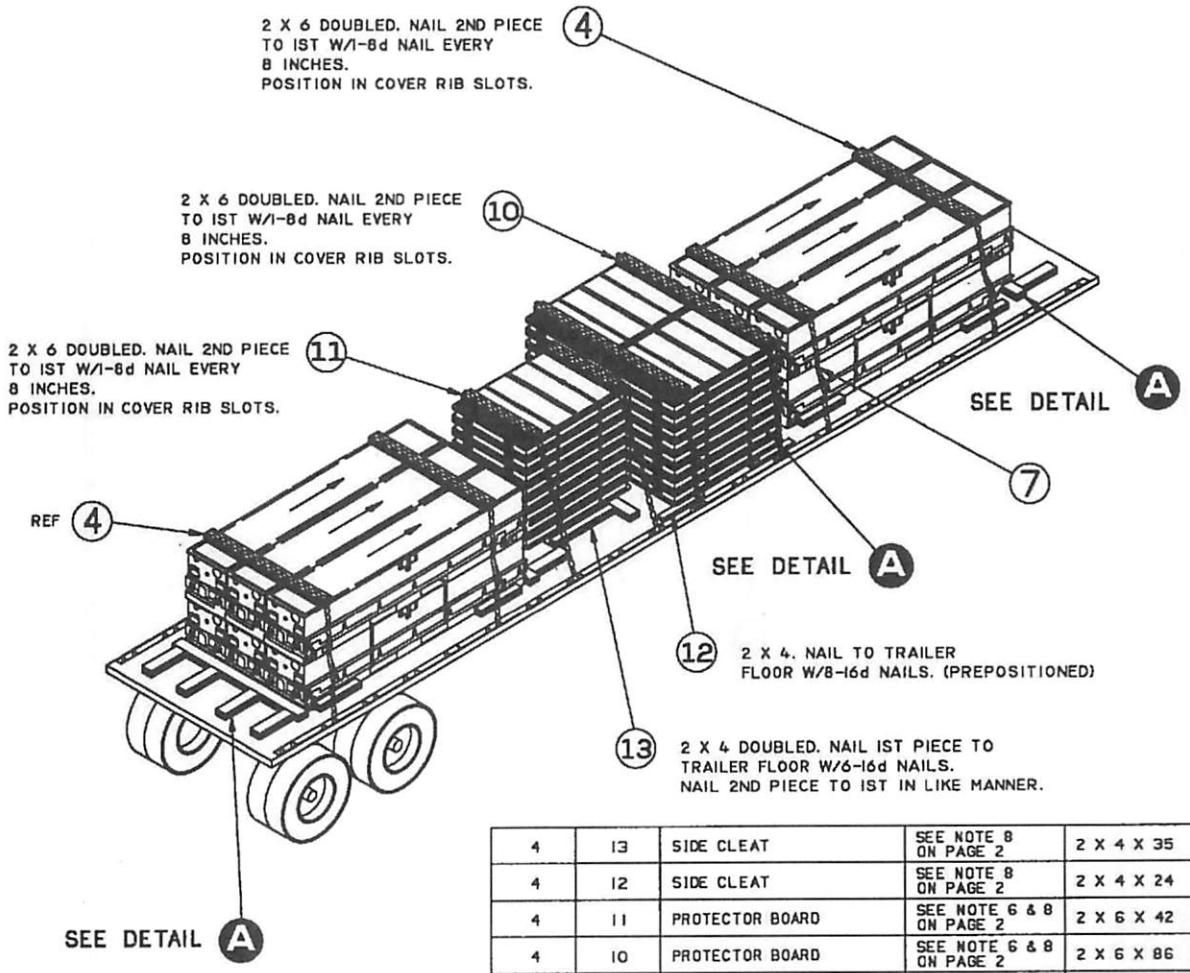
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NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY.

NOTE

DIRECTION OF ARROW INDICATES FORWARD END OF CONTAINER



TRUCKLOAD DATA

40 FT THROUGH 48 FT TRAILER

NUMBER OF CNU-443A/E CONTAINERS. 12

NUMBER OF CNU-450/E CONTAINERS 24

WEIGHT . . . . . 19284 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

LIST OF MATERIALS

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	13	SIDE CLEAT	SEE NOTE 8 ON PAGE 2	2 X 4 X 35
4	12	SIDE CLEAT	SEE NOTE 8 ON PAGE 2	2 X 4 X 24
4	11	PROTECTOR BOARD	SEE NOTE 6 & 8 ON PAGE 2	2 X 6 X 42
4	10	PROTECTOR BOARD	SEE NOTE 6 & 8 ON PAGE 2	2 X 6 X 86
5	9	DECK BOARD	SEE NOTE 8 ON PAGE 2	1 X 6 X 35
2	8	SKID	SEE NOTE 8 ON PAGE 2	4 X 4 X 44
8	7	CHAIN AND LOAD BINDER	SEE NOTE 5 ON PAGE 2	5/16 OR 3/8
4	6	VERTICAL	SEE NOTE 6 ON PAGE 2	2 X 6 X 24
4	5	HORIZONTAL	SEE NOTE 6 ON PAGE 2	2 X 6 X 19 1/2
12	4	PROTECTOR BOARD	SEE NOTE 6 & 8 ON PAGE 2	2 X 6 X 68 1/2
16	3	SIDE CLEAT	SEE NOTE 6 ON PAGE 2	2 X 4 X 24
32	2	BACKUP CLEAT	SEE NOTE 8 ON PAGE 2	2 X 6 X 24
24	1	CROSSMEMBER	SEE NOTE 8 ON PAGE 2	2 X 6 X 12

FTL 40 FT THROUGH 48 FT TRAILER (FLATBED)
(40 FT TRAILER SHOWN)

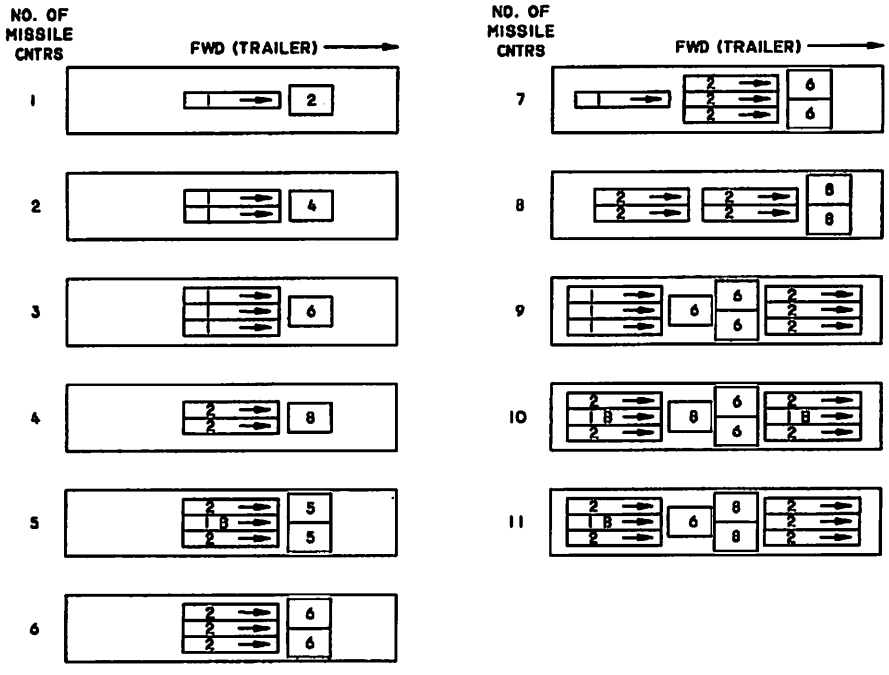
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LTL 40 FT THROUGH 48 FT TRAILERS (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN, FOR THE NUMBER OF MISSILE CONTAINERS AND WING AND CANARD CONTAINERS TO BE SHIPPED, FROM TABLE I.
2. POSITION UNIT LOADS ON THE TRAILER AS SHOWN IN TABLE I. BLOCK AND TIE DOWN USING THE PRINCIPLES SHOWN ON PAGES 3 AND 5 AND THE GENERAL DOCUMENT MIL-STD-1320 (NAVY).
3. THE LTL PLAN ON PAGE 5 SHOWS THE LESS-THAN-TRUCKLOAD METHODS FOR STACKS OF ONE AND TWO CONTAINERS HIGH AND THE APPLICATION OF DETAIL B SPACER/HOLD DOWN.
4. THE NUMBERS IN EACH ROW OF A STACK IS THE NUMBER OF COTAINERS STACKED IN THE ROW. A "B" BEFORE THE NUMBER REQUIRES THE USE OF DETAIL B SPACERS ON THE PROTECTOR BOARD ASSEMBLIES.

TABLE I

LOAD PATTERN *



* LOAD PLAN SHOWS FLOOR PLAN FOR TRAILER.
ARROWS INDICATE FORWARD END OF CONTAINER.

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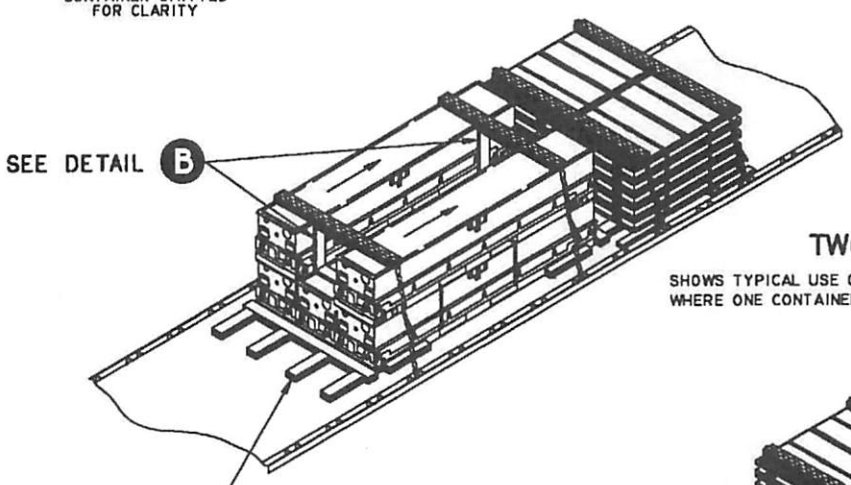
TYPICAL CONSTRUCTION FOR STACKS OF ONE AND TWO LAYERS

NOTE

TARPAULINS OVER CONTAINER OMITTED FOR CLARITY

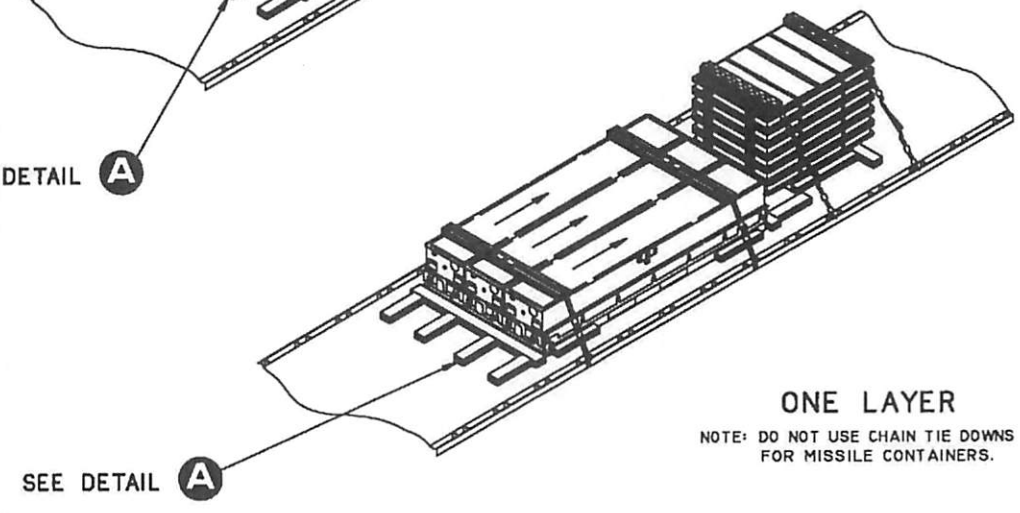
NOTE

DIRECTION OF ARROW INDICATES FORWARD END OF CONTAINER



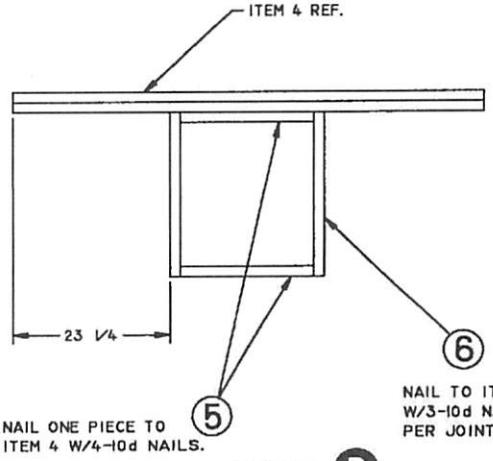
TWO LAYERS

SHOWS TYPICAL USE OF DETAIL B SPACER/HOLD DOWNS WHERE ONE CONTAINER IS OMITTED FROM THE STACK.



ONE LAYER

NOTE: DO NOT USE CHAIN TIE DOWNS FOR MISSILE CONTAINERS.



DETAIL B

SPACER/HOLD DOWN ASSEMBLY (2 REQUIRED PER STACK WHEN NEEDED)

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UNITIZING PROCEDURE

FOR MISSILE CONTAINERS (CNU-443A/E)

1. USING A FORK TRUCK, STACK ONE CONTAINER ON TOP OF THE FIRST CONTAINER. BE SURE TO ALIGN THE STACKING FEATURES.
2. POSITION TWO 1 1/4 X .035 STEEL STRAPS THROUGH THE FORK POCKETS OF THE CONTAINERS, AT THE ENDS OF THE POCKETS, AS SHOWN. TENSION AND SEAL WITH ONE DOUBLE NOTCHED SEAL (SEE NOTE 8 ON PAGE 2).

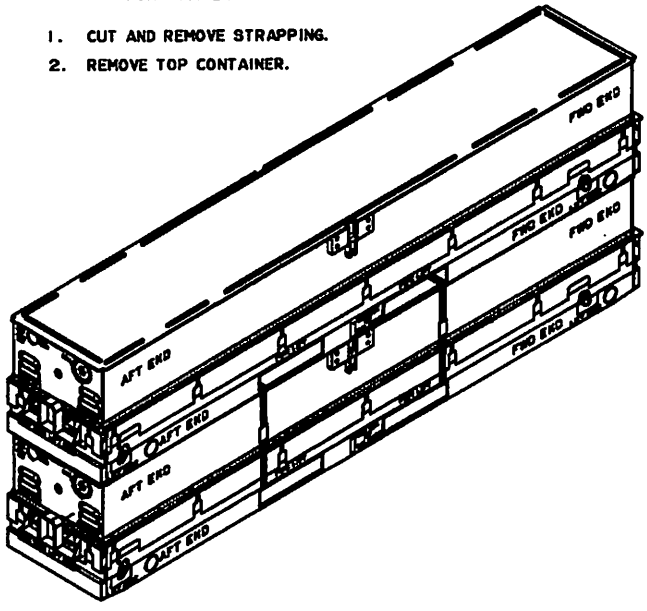
WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED CONTAINERS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPRIATE BACKUP OF OUTBOARD CONTAINERS.

DEUNITIZING PROCEDURE

FOR MISSILE CONTAINERS (CNU-443A/E)

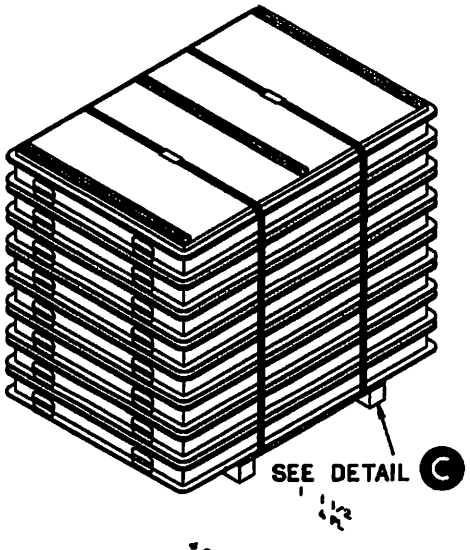
1. CUT AND REMOVE STRAPPING.
2. REMOVE TOP CONTAINER.



UNITIZING PROCEDURE

FOR WING AND CANARD CONTAINERS (CNU-450/E)

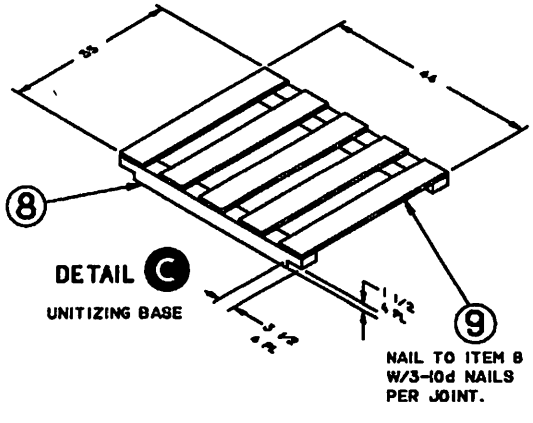
1. STACK TWO, FOUR, SIX, OR EIGHT CONTAINERS CENTERED ON THE UNITIZING BASE.
   NOTE: TWO WING AND CANARD CONTAINERS ARE REQUIRED FOR EACH MISSILE CONTAINER.
2. POSITION TWO 1 1/4 X .035 STEEL STRAPS UNDER THE UNITIZING BASE AND OVER THE CONTAINERS, AS SHOWN. TENSION AND SEAL WITH ONE DOUBLE NOTCHED SEAL (SEE NOTE 8 ON PAGE 2). .



DEUNITIZING PROCEDURE

FOR WING AND CANARD CONTAINERS (CNU-450/E)

1. CUT AND REMOVE STRAPPING.
2. REMOVE CONTAINERS.



REVIEW ACTIVITY:
NAVY-AS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N1037)

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